

TUCSON 11 W, AZ, USA

WMO: 722750

Lat: 32.2395N

Lon: 111.1696W

Elev: 833

StdP: 91.71

Time Zone: -7.00 (W07)

Period: 02-19

WBAN: 53131

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.1	1.8	-19.4	0.7	20.0	-17.1	0.9	20.4	5.0	13.2	4.4	13.6	0.9	N/A	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	15.7	40.3	17.9	39.1	17.8	38.0	17.7	22.1	29.7	21.7	29.7	21.2	29.8	2.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.4	16.7	24.4	19.7	16.0	24.6	18.9	15.1	25.2	69.1	29.5	67.3	29.4	65.6	29.4	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.2	3.8	3.3	-3.4	42.5	2.0	1.1	-4.9	43.3	-6.0	44.0	-7.2	44.6	-8.6	45.4		
			-5.6	23.3	2.8	0.7	-7.6	23.8	-9.2	24.2	-10.8	24.6	-12.9	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	12.1	13.0	16.7	19.8	24.1	29.9	30.7	29.6	27.6	22.2	16.7	11.6
	DBStd	7.71	3.79	3.99	3.96	3.62	3.62	2.60	2.42	2.27	2.60	3.63	4.11	3.89
	HDD10.0	74	22	16	3	1	0	0	0	0	0	0	4	29
	HDD18.3	750	194	153	77	26	3	0	0	0	0	11	77	209
	CDD10.0	4169	88	99	211	296	438	598	641	608	527	379	205	79
	CDD18.3	1803	1	3	27	71	183	348	382	350	277	132	28	1
	CDH23.3	25149	18	79	443	1032	2639	5527	5447	4694	3463	1474	312	19
CDH26.7	14102	0	9	115	384	1401	3593	3314	2721	1879	616	69	0	
Wind	WSAvg	1.6	1.5	1.5	1.6	1.7	1.8	1.7	1.5	1.3	1.5	1.5	1.6	1.4
Precipitation	PrecAvg	272	25	23	21	7	5	5	48	52	29	14	14	29
	PrecMax	469	131	95	78	33	31	49	129	127	68	97	51	90
	PrecMin	146	0	0	1	0	0	0	6	14	5	1	1	0
	PrecStd	77	30	23	19	8	7	10	27	26	15	18	13	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	28.1	31.8	34.2	38.3	42.3	41.7	40.7	38.6	35.2	30.7	25.5
		MCWB	9.9	10.9	12.2	13.4	14.9	17.2	19.0	19.4	18.0	15.7	12.2	10.3
	2%	DB	23.4	25.9	29.6	32.3	36.5	40.4	40.1	39.1	37.2	33.4	28.7	23.4
		MCWB	9.7	10.2	11.6	12.7	14.2	16.7	19.0	19.7	18.0	15.0	12.4	9.9
	5%	DB	21.6	23.7	27.7	30.7	35.0	39.3	38.9	37.8	36.0	31.9	26.9	21.3
		MCWB	8.9	9.6	10.8	12.1	13.9	16.4	19.0	19.3	17.9	14.8	12.1	9.5
	10%	DB	19.6	21.4	25.9	29.0	33.6	38.1	37.5	36.4	34.6	30.4	25.0	19.2
		MCWB	8.2	9.1	10.4	11.6	13.2	16.0	19.1	19.3	17.9	14.3	11.6	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	13.4	13.0	14.5	15.8	20.7	22.8	22.9	21.8	19.9	14.9	12.9
		MCDB	18.4	17.3	25.7	30.9	32.8	30.9	29.2	30.5	28.7	23.2	24.2	18.7
	2%	WB	11.6	12.2	12.2	13.5	15.0	19.9	22.1	22.3	21.1	18.0	13.8	11.7
		MCDB	17.6	18.7	24.9	30.0	33.6	31.1	29.4	30.2	28.5	26.2	23.5	19.2
	5%	WB	10.6	11.1	11.5	12.9	14.1	19.0	21.7	21.8	20.6	16.9	13.0	10.8
		MCDB	17.7	20.1	24.3	28.7	32.6	32.3	29.6	30.1	28.4	26.7	23.3	18.3
	10%	WB	9.6	10.3	10.8	12.1	13.5	18.0	21.3	21.3	20.1	16.0	12.2	9.9
		MCDB	17.7	20.4	23.7	27.7	31.2	33.7	29.8	30.2	28.3	26.4	22.9	17.4
Mean Daily Temperature Range	5% DB	MDBR	12.7	13.3	14.6	15.3	15.7	12.4	12.4	12.8	13.8	13.4	12.4	
		MCDBR	14.9	16.7	17.0	17.3	17.2	16.3	13.9	14.3	14.3	15.1	15.3	14.9
		MCWBR	6.8	7.1	6.7	6.7	6.1	5.2	3.5	3.6	4.2	5.4	6.1	6.6
	5% WB	MCDBR	11.5	13.6	15.0	16.1	16.3	13.1	11.4	11.7	11.3	11.8	13.2	12.2
	MCWBR	5.8	6.1	5.9	6.1	5.7	4.1	2.7	3.0	3.4	4.5	5.7	5.8	
Clear-Sky Solar Irradiance	taub	0.260	0.264	0.283	0.296	0.306	0.327	0.385	0.372	0.332	0.295	0.269	0.259	
	taud	2.636	2.628	2.541	2.487	2.492	2.440	2.409	2.434	2.516	2.571	2.603	2.631	
	Ebn at Noon	985	1009	1003	994	978	952	896	906	936	957	963	968	
	Edh at Noon	75	83	98	108	109	115	118	113	100	87	77	71	
All-Sky Solar Radiation	RadAvg	3.59	4.54	6.02	7.40	8.20	8.26	6.94	6.53	5.97	5.06	3.97	3.28	
	RadStd	0.25	0.35	0.28	0.22	0.28	0.42	0.40	0.22	0.27	0.30	0.19	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+181	N/A
(47)	Regional (14 neighbors)	+0.27	N/A	N/A	N/A	N/A	N/A	N/A	-75	+104	N/A

Nomenclature: See separate page